

ML Specification Report — us-patent-phrase-to-phrase-matching

Patent Phrase Semantic-Similarity Scoring (NLP pair regression) • our from-scratch agent (DeBERTa-v3 fine-tunes + NNLS blend — deep-learning fallback lane)

Figures grounded in the competition's `facts.json` (built by `benchmark_infra/collect_dl_facts.py` from `config.yaml`, `experiments.json`, `scripts/*_cv.json`, `scripts/ensemble_run2_decision.json`, `STATUS.md`, `headless_run2.log`), with grading figures from the offline MLE-bench records in `benchmark_results/run2/`. This is a special (deep-learning fallback) lane: grading is offline MLE-bench against a held-out private split — there is no live Kaggle leaderboard, and the framework's DL fields (backbone, learning-rate schedule, batch size, epochs, transfer learning) are native here rather than translated from GBDT.

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Overview

The task is to score the semantic similarity of a pair of patent phrases (`anchor` , `target`) within a CPC classification context, against human-rated labels in $[0, 1]$; the metric is the **Pearson correlation coefficient (maximize)**. Pair-similarity NLP is outside the GBDT tree-search pipeline's home turf, so the agent fell back to fine-tuning pretrained transformer encoders. The champion is an **NNLS blend of four members — deberta-v3-large at two seeds, deberta-v3-base, and a LightGBM lexical baseline** — scoring **honest leave-fold-out Pearson r 0.83184** (full-OOF 0.83206) and grading **0.86658** on the offline MLE-bench private split: a **silver medal** (silver threshold 0.863, gold 0.87). The NVIDIA reproduce-agent grades 0.8616 (bronze, exactly on its threshold) and AIDE 0.65821 (no medal) — **our agent wins the three-way comparison**.

Why it matters. Patent examination hinges on deciding whether a claimed phrase means the same thing as prior-art phrasing — "television set" vs "TV set", but judged inside a technical domain.

Automated context-aware similarity scoring helps examiners and inventors search a corpus of millions of documents where verbatim matching fails.

Data

Purpose of Data. Each row is a phrase pair — an `anchor` phrase, a `target` phrase, and a CPC context code — with a human-rated similarity `score` in $[0, 1]$ to regress. **Data Format** is **tabular CSV of short text fields** plus a categorical context code. **Data Volume:** the graded test split has **3,648 pairs** (the deliverable submission's verified row count); the train-split row count was **not recorded** in this lane's structured records (no `eda_summary.json` was produced under the fallback's time budget).

Data Quality. Missing-value / duplicate profiling was **not recorded**. The defining wrinkle, established by the run's own honesty check, is the **train/test anchor overlap**: test anchors are 100% seen in train (the pairs are unseen), so a GroupKFold-by-anchor CV measures a strictly harder generalization problem than the graded one — the run recorded "true score likely $\geq$ CV", and the offline grade confirmed it (0.86658 graded vs 0.83184 CV).

Annotation Guidelines. The label is a human-rated similarity score in $[0, 1]$; a submission is a real-valued prediction per pair, scored by Pearson correlation — a pure linear-association metric, so raw (clipped) scores are submitted with no thresholding or calibration.

Feature Set. Two representations feed the members:

Group	Representation
Transformer input (3 DeBERTa members)	<code>anchor [SEP] target [SEP] <CPC section title> <context code></code> , <code>max_len</code> 96
Lexical features (LGBM baseline, 14 features)	<code>cos_char</code> , <code>cos_word</code> , <code>jaccard</code> , <code>n_common</code> , <code>len_a</code> , <code>len_t</code> , <code>nw_a</code> , <code>nw_t</code> , <code>len_ratio</code> , <code>a_in_t</code> , <code>t_in_a</code> , <code>same_first_word</code> , <code>context_code</code> , <code>section</code>

Splitting strategy. The canonical scheme is **GroupKFold(4) grouped by anchor** — chosen because anchors repeat across many rows, and grouping prevents the model from scoring an anchor it has memorized. GroupKFold is seed-independent, so all transformer OOFs share identical folds and every blend score is computed on a consistent partition. The LGBM baseline used a StratifiedKFold(5) on score (seed 42) — a non-grouped, optimistic solo estimate, honestly flagged in the records; it still earns blend weight on the grouped-DeBERTa folds. **LB status:** offline MLE-bench grade only — no live public/private leaderboard.

Models & Architecture

Purpose of Architecture. Maximize Pearson r between predicted and human similarity scores.

Architecture Type is **fine-tuned pretrained transformer encoders + a lexical GBDT, combined by NNLS** — the DL fields are native. The champion is the **4-member NNLS blend** `nnls_all`, with weights:

Member	Backbone	Seed	Solo OOF r	NNLS weight
deberta_large_s1337	microsoft/deberta-v3-large	1337	0.8213	0.3659
deberta_large_s42	microsoft/deberta-v3-large	42	0.8197	0.2618
deberta_base_s42	microsoft/deberta-v3-base	42	0.8060	0.2531
lgbm lexical baseline	LightGBM (14 features)	42	0.6113	0.1192

Input Format / Dimension. Transformer members consume the tokenized pair-plus-context string truncated to **max_len 96**; the LGBM member consumes a 14-dimensional lexical feature vector.

Architecture Description. Each DeBERTa member is the pretrained encoder with a **mean-pooling head** over token embeddings and a **BCEWithLogits** objective on the [0, 1] score (sigmoid output). The blend layer is a **non-negative least squares (NNLS)** fit on OOF predictions, with the winning candidate chosen by **honest leave-fold-out refit** (weights refit on 3 folds, scored on the held-out fold). **Model Complexity:** parameter counts were **not recorded**; the members are fully identified by their published backbone names.

Training procedures

Training proceeded as a **member ladder plus a blend-candidate search** (tree search deliberately skipped — see below), all scored as OOF Pearson r:

Stage	Configuration	OOF r
1	LGBM char/word TF-IDF + lexical (safety baseline)	0.6113
2	deberta-v3-base, seed 42	0.8060
3	deberta-v3-large, seed 42	0.8197
4	deberta-v3-large, seed 1337 (seed bagging)	0.8213
5	NNLS blend of all 4 (honest leave-fold-out)	0.83184

The blend stage evaluated 9 candidates by honest score: solo members (above), `mean_larges` 0.82598, `mean_all_deberta` 0.83015, `nnls_larges` 0.82597, `nnls_deberta` 0.83003, and the winner **`nnls_all` 0.83184** (full-OOF 0.83206).

The **Loss Function** is **BCEWithLogitsLoss** on the [0, 1] score, with model selection on Pearson r. The **Optimization Algorithm** is **fused AdamW** (weight decay 0.01); blend weights by NNLS. The **Learning**

Rate (native) is **1e-5 (large) / 2e-5 (base)** for the encoder body with a separate **head LR of 0.0001**. The **Learning Rate Scheduler** (native) is **cosine decay with 10% warmup** — the recipe from the project's deberta-v3-large collapse memory, smoke-probed before committing (val_r 0.64 @ 40 steps) and delivering 8/8 clean large folds with zero collapses. The **Batch Size** (native) is **32**, for **3 epochs** per fold, under **bf16 autocast over fp32 master weights**.

Training Duration: ~2.5 h for the whole run; a 4-fold deberta-v3-large fine-tune takes ~35 min. **Training Memory: not recorded**.

Transfer Learning (native DL field): **the load-bearing ingredient** — `microsoft/deberta-v3-base` and `microsoft/deberta-v3-large` pretrained checkpoints, explicitly allowed by the rules (`pretrained_models_allowed: true`); this is the run's main source of injected prior knowledge. **Data Augmentation** (native DL field): **none recorded**; the variance-reduction analogue actually used is **transformer seed bagging** (the same large recipe at seeds 42 and 1337, solos 0.8197 / 0.8213).

Reproducibility Standards. Fixed seeds (42 / 1337); GroupKFold's seed-independence guarantees identical folds across all members; all training jobs were foreground-waited (no orphaned background runs). The honest-vs-full-OOF gap of the chosen blend is only 0.0002, evidence the weight fitting is not overfitting the OOF. **Tree search was deliberately skipped:** each meaningful node is a ~35-min transformer fine-tune, so a ~60-node search is ~35 GPU-hours against the ~4 h budget; the light blend-level candidate search (9 candidates, honest scoring) served as the Stage-4 loop.

Inference procedures

Decision Threshold is N/A — Pearson r is a correlation metric; raw real-valued predictions are submitted. The only post-processing is a **clip to [0, 1]** (the label's range). The submission (`submissions/ensemble_run2_nnls_all.csv` , promoted to `submission.csv`) has 3,648 rows, format-verified against `sample_submission.csv` . **Inference Duration / Inference Memory: not recorded**.

Evaluation & Benchmarking

Task for Performance Evaluation. Predict test pairs' similarity scores. **Performance Metrics.** Pearson r (maximize): 0.6113 → 0.8060 → 0.8197 → 0.8213 → **0.83184** across the five stages; offline MLE-bench grade **0.86658** against thresholds gold 0.87 / silver 0.863 / bronze 0.8616 / median 0.851 — a **silver medal**, above median. Versus the archived run1 (OOF 0.8158, offline grade 0.8575), run2 gains +0.016 OOF, driven by the large backbone under the stable recipe plus seed bagging. The CV-to-grade gap (0.83184 → 0.86658) is consistent with the recorded anchor-overlap analysis: the grouped CV is conservative relative to the graded distribution.

Performance Benchmarking. Grading is **offline MLE-bench** (all three agents scored on the same held-out private split — not a live leaderboard):

Agent	Approach	Pearson r (offline grade)	Medal
Our agent	from-scratch DeBERTa-v3 fine-tunes + NNLS blend	0.86658	silver
NVIDIA	(approach not recorded in the grade record)	0.8616	bronze
AIDE	(approach not recorded in the grade record)	0.65821	none

Our agent is the only silver in the three-way; NVIDIA lands exactly on the bronze threshold (0.8616), and AIDE finishes below median. The honest caveats: the comparison is grade-only (the comparators' pipelines are not described in the run2 grade records), and our margin over NVIDIA rests on the honest blend rather than any single member — the best solo (0.8213 CV) would not have been separated from a well-executed reproduction.

Fields marked "not recorded": train-split volume and missing/duplicate profiling (no EDA summary artifact); model parameter counts; training peak memory; inference duration and memory; NVIDIA/AIDE approach details. Grading is offline MLE-bench — no public/private live leaderboard exists for this lane.